

The Bivariate Normal CEF

Why the CEF is Linear Under Joint Normality

PLSC 30700: Estimation I

1 Setup

Suppose two random variables are jointly bivariate normal:

$$\begin{pmatrix} X \\ Y \end{pmatrix} \sim N\left(\begin{pmatrix} \mu_X \\ \mu_Y \end{pmatrix}, \begin{pmatrix} \sigma_X^2 & \rho \sigma_X \sigma_Y \\ \rho \sigma_X \sigma_Y & \sigma_Y^2 \end{pmatrix}\right)$$

where $\rho = \text{Cov}(X, Y)/(\sigma_X \sigma_Y)$ is the correlation. Our goal is to derive the conditional distribution $f_{Y|X}(y | x)$ and show that $\mathbb{E}[Y | X = x]$ is a *linear* function of x . This is one of the classical motivations for linear regression.

Throughout, we write densities as functions of realized values (x, y) . Define the standardized values

$$x^* = \frac{x - \mu_X}{\sigma_X}, \quad y^* = \frac{y - \mu_Y}{\sigma_Y}.$$

(These are numbers, not random variables. The corresponding random variables would be $X^* = (X - \mu_X)/\sigma_X$ and $Y^* = (Y - \mu_Y)/\sigma_Y$, but inside density expressions we always evaluate at a specific point.)

2 The three densities we need

2.1 Marginal density of X

Since $X \sim N(\mu_X, \sigma_X^2)$:

$$f_X(x) = \frac{1}{\sqrt{2\pi} \sigma_X} \exp\left(-\frac{1}{2}(x^*)^2\right) \quad (1)$$

2.2 Joint density of (X, Y)

The bivariate normal density is:

$$f_{X,Y}(x, y) = \frac{1}{2\pi \sigma_X \sigma_Y \sqrt{1 - \rho^2}} \exp\left(-\frac{1}{2(1 - \rho^2)} \left[(x^*)^2 - 2\rho x^* y^* + (y^*)^2\right]\right) \quad (2)$$

2.3 Conditional density: the ratio

By definition:

$$f_{Y|X}(y | x) = \frac{f_{X,Y}(x, y)}{f_X(x)}$$

We will compute this ratio and simplify.

3 Step 1: Form the ratio

Dividing (2) by (1):

$$f_{Y|X}(y | x) = \frac{\frac{1}{2\pi \sigma_X \sigma_Y \sqrt{1 - \rho^2}} \exp\left(-\frac{(x^*)^2 - 2\rho x^* y^* + (y^*)^2}{2(1 - \rho^2)}\right)}{\frac{1}{\sqrt{2\pi} \sigma_X} \exp\left(-\frac{(x^*)^2}{2}\right)}$$

Simplify the constant out front. The ratio of the leading fractions is:

$$\frac{1/(2\pi \sigma_X \sigma_Y \sqrt{1 - \rho^2})}{1/(\sqrt{2\pi} \sigma_X)} = \frac{\sqrt{2\pi} \sigma_X}{2\pi \sigma_X \sigma_Y \sqrt{1 - \rho^2}} = \frac{1}{\sqrt{2\pi} \sigma_Y \sqrt{1 - \rho^2}}$$

Simplify the exponential. Using $e^a/e^b = e^{a-b}$, the exponent becomes:

$$\begin{aligned} & -\frac{(x^*)^2 - 2\rho x^* y^* + (y^*)^2}{2(1 - \rho^2)} - \left(-\frac{(x^*)^2}{2}\right) \\ &= -\frac{(x^*)^2 - 2\rho x^* y^* + (y^*)^2}{2(1 - \rho^2)} + \frac{(x^*)^2}{2} \end{aligned}$$

Put the second term over the common denominator $2(1 - \rho^2)$:

$$\begin{aligned} &= -\frac{(x^*)^2 - 2\rho x^* y^* + (y^*)^2}{2(1 - \rho^2)} + \frac{(1 - \rho^2)(x^*)^2}{2(1 - \rho^2)} \\ &= \frac{-(x^*)^2 + 2\rho x^* y^* - (y^*)^2 + (x^*)^2 - \rho^2(x^*)^2}{2(1 - \rho^2)} \end{aligned}$$

The $(x^*)^2$ terms cancel:

$$= \frac{2\rho x^* y^* - (y^*)^2 - \rho^2(x^*)^2}{2(1 - \rho^2)}$$

Factor out a negative sign:

$$= -\frac{(y^*)^2 - 2\rho x^* y^* + \rho^2(x^*)^2}{2(1 - \rho^2)}$$

4 Step 2: Recognize the perfect square

The numerator in the exponent is:

$$(y^*)^2 - 2\rho x^* y^* + \rho^2 (x^*)^2 = (y^* - \rho x^*)^2$$

You can verify this by expanding the right side:

$$(y^* - \rho x^*)^2 = (y^*)^2 - 2\rho x^* y^* + \rho^2 (x^*)^2 \quad \checkmark$$

So the exponent simplifies to:

$$-\frac{(y^* - \rho x^*)^2}{2(1 - \rho^2)}$$

Putting it all together:

$$f_{Y|X}(y | x) = \frac{1}{\sqrt{2\pi} \sigma_Y \sqrt{1 - \rho^2}} \exp\left(-\frac{(y^* - \rho x^*)^2}{2(1 - \rho^2)}\right) \quad (3)$$

5 Step 3: Rewrite in terms of y and x

Substitute $y^* = (y - \mu_Y)/\sigma_Y$ and $x^* = (x - \mu_X)/\sigma_X$ back in:

$$y^* - \rho x^* = \frac{y - \mu_Y}{\sigma_Y} - \rho \frac{x - \mu_X}{\sigma_X} = \frac{1}{\sigma_Y} \left[y - \mu_Y - \rho \frac{\sigma_Y}{\sigma_X} (x - \mu_X) \right]$$

So the exponent in (3) becomes:

$$-\frac{1}{2(1 - \rho^2)} \left(\frac{y - \mu_Y - \rho \frac{\sigma_Y}{\sigma_X} (x - \mu_X)}{\sigma_Y} \right)^2 = -\frac{1}{2} \left(\frac{y - \mu_Y - \rho \frac{\sigma_Y}{\sigma_X} (x - \mu_X)}{\sigma_Y \sqrt{1 - \rho^2}} \right)^2$$

The full conditional density is therefore:

$$f_{Y|X}(y | x) = \frac{1}{\sqrt{2\pi} \sigma_Y \sqrt{1 - \rho^2}} \exp\left(-\frac{1}{2} \left(\frac{y - \left[\mu_Y + \rho \frac{\sigma_Y}{\sigma_X} (x - \mu_X) \right]}{\sigma_Y \sqrt{1 - \rho^2}} \right)^2\right) \quad (4)$$

6 Step 4: Read off the conditional distribution

Equation (4) is the density of a normal distribution. Recall that $N(\mu, \tau^2)$ has density

$$\frac{1}{\sqrt{2\pi} \tau} \exp\left(-\frac{(y - \mu)^2}{2\tau^2}\right).$$

Matching terms:

Conditional distribution: $Y \mid X = x \sim N(\mu_{Y|X}, \sigma_{Y|X}^2)$

• **Conditional mean:** $\mathbb{E}[Y \mid X = x] = \mu_Y + \rho \frac{\sigma_Y}{\sigma_X} (x - \mu_X)$

• **Conditional variance:** $\text{Var}(Y \mid X = x) = \sigma_Y^2(1 - \rho^2)$

7 Step 5: Connect to regression

7.1 The CEF is linear

The conditional expectation function is:

$$\mathbb{E}[Y \mid X = x] = \underbrace{\mu_Y - \rho \frac{\sigma_Y}{\sigma_X} \mu_X}_{\text{intercept}} + \underbrace{\rho \frac{\sigma_Y}{\sigma_X}}_{\text{slope}} \cdot x$$

This is a linear function of x . Under joint normality, the CEF is *exactly* linear—no approximation involved.

7.2 The slope is the regression coefficient

Recall that the population regression coefficient from the BLP is:

$$\beta = \frac{\text{Cov}(Y, X)}{\text{Var}(X)}$$

Since $\text{Cov}(Y, X) = \rho \sigma_X \sigma_Y$ and $\text{Var}(X) = \sigma_X^2$:

$$\frac{\text{Cov}(Y, X)}{\text{Var}(X)} = \frac{\rho \sigma_X \sigma_Y}{\sigma_X^2} = \rho \frac{\sigma_Y}{\sigma_X}$$

This is exactly the slope of the conditional mean. Under joint normality, the BLP coefficient and the CEF slope coincide.

7.3 The conditional variance is constant

$\text{Var}(Y \mid X = x) = \sigma_Y^2(1 - \rho^2)$ does not depend on x . This is **homoskedasticity**: the spread of Y around its conditional mean is the same for every value of X .

Under joint normality, the key ingredients of the classical linear-normal regression setup follow automatically: a linear CEF, conditionally normal errors, and homoskedasticity.

7.4 The error variance

The regression error is $e = Y - \mathbb{E}[Y | X]$. By construction, $\mathbb{E}[e | X] = 0$. Because the conditional variance is constant under joint normality:

$$\text{Var}(e | X) = \sigma_Y^2(1 - \rho^2)$$

and therefore

$$\text{Var}(e) = \mathbb{E}[\text{Var}(e | X)] = \sigma_Y^2(1 - \rho^2) = \sigma_Y^2 - \frac{[\text{Cov}(Y, X)]^2}{\text{Var}(X)}.$$

(In general, $\text{Var}(Y | X)$ is a random variable measurable with respect to X , but here it equals a constant, so $\text{Var}(e) = \text{Var}(e | X)$.)

When $|\rho|$ is close to 1, the errors are small and regression explains most of the variation in Y . When $\rho = 0$, the conditional variance equals the unconditional variance σ_Y^2 —knowing X tells us nothing about Y .

8 Does this break down in higher dimensions?

No. Everything above generalizes from one regressor to k regressors with the same structure. The algebra uses block matrix operations instead of scalar fractions, but the logic is identical: partition, condition, complete the square.

8.1 Setup

Suppose $(Y, \mathbf{X})$ are jointly multivariate normal, where Y is a scalar and $\mathbf{X}$ is a $k \times 1$ vector. Partition the mean and covariance:

$$\begin{pmatrix} Y \\ \mathbf{X} \end{pmatrix} \sim N\left(\begin{pmatrix} \mu_Y \\ \boldsymbol{\mu}_X \end{pmatrix}, \begin{pmatrix} \sigma_Y^2 & \boldsymbol{\Sigma}_{YX} \\ \boldsymbol{\Sigma}_{XY} & \boldsymbol{\Sigma}_{XX} \end{pmatrix}\right)$$

where $\boldsymbol{\Sigma}_{YX} = \text{Cov}(Y, \mathbf{X})$ is $1 \times k$ and $\boldsymbol{\Sigma}_{XX} = \text{Var}(\mathbf{X})$ is $k \times k$.

8.2 The multivariate normal conditional (statement)

The conditional distribution is:

$$Y | \mathbf{X} = \mathbf{x} \sim N(\mu_{Y|\mathbf{X}}, \sigma_{Y|\mathbf{X}}^2)$$

- **Conditional mean:** $\mathbb{E}[Y | \mathbf{X} = \mathbf{x}] = \mu_Y + \boldsymbol{\Sigma}_{YX} \boldsymbol{\Sigma}_{XX}^{-1}(\mathbf{x} - \boldsymbol{\mu}_X)$
- **Conditional variance:** $\text{Var}(Y | \mathbf{X} = \mathbf{x}) = \sigma_Y^2 - \boldsymbol{\Sigma}_{YX} \boldsymbol{\Sigma}_{XX}^{-1} \boldsymbol{\Sigma}_{XY}$

8.3 Verifying the bivariate case

When $k = 1$, $\Sigma_{XX} = \sigma_X^2$, $\Sigma_{YX} = \text{Cov}(Y, X) = \rho \sigma_Y \sigma_X$, and $\Sigma_{XX}^{-1} = 1/\sigma_X^2$. Substituting:

$$\mathbb{E}[Y \mid X = x] = \mu_Y + \frac{\rho \sigma_Y \sigma_X}{\sigma_X^2}(x - \mu_X) = \mu_Y + \rho \frac{\sigma_Y}{\sigma_X}(x - \mu_X) \quad \checkmark$$

$$\text{Var}(Y \mid X = x) = \sigma_Y^2 - \frac{(\rho \sigma_Y \sigma_X)^2}{\sigma_X^2} = \sigma_Y^2(1 - \rho^2) \quad \checkmark$$

8.4 Derivation sketch

The multivariate derivation follows exactly the same three moves as the bivariate case:

Step 1: Form the ratio. Write the joint density of $(Y, \mathbf{X})$ and divide by the marginal density of $\mathbf{X}$. Both are multivariate normal densities, so the ratio is an exponential of a difference of quadratic forms:

$$f_{Y|\mathbf{X}}(y \mid \mathbf{x}) \propto \exp\left(-\frac{1}{2} \left[\begin{pmatrix} y - \mu_Y \\ \mathbf{x} - \boldsymbol{\mu}_X \end{pmatrix}' \Sigma^{-1} \begin{pmatrix} y - \mu_Y \\ \mathbf{x} - \boldsymbol{\mu}_X \end{pmatrix} - (\mathbf{x} - \boldsymbol{\mu}_X)' \Sigma_{XX}^{-1} (\mathbf{x} - \boldsymbol{\mu}_X) \right]\right)$$

The key tool is the **block inverse formula**. For a partitioned positive-definite matrix:

$$\Sigma^{-1} = \begin{pmatrix} \sigma_Y^2 & \Sigma_{YX} \\ \Sigma_{XY} & \Sigma_{XX} \end{pmatrix}^{-1} = \begin{pmatrix} \omega^{-1} & -\omega^{-1} \Sigma_{YX} \Sigma_{XX}^{-1} \\ -\Sigma_{XX}^{-1} \Sigma_{XY} \omega^{-1} & \Sigma_{XX}^{-1} + \Sigma_{XX}^{-1} \Sigma_{XY} \omega^{-1} \Sigma_{YX} \Sigma_{XX}^{-1} \end{pmatrix}$$

where $\omega = \sigma_Y^2 - \Sigma_{YX} \Sigma_{XX}^{-1} \Sigma_{XY}$ is the *Schur complement*—the conditional variance.

Step 2: Complete the square. Substitute the block inverse and expand. All terms involving $\mathbf{x}$ alone (without y) cancel between the joint and marginal quadratic forms—exactly as $(x^*)^2$ cancelled in the bivariate case. What remains is:

$$-\frac{1}{2\omega} \left(y - \mu_Y - \Sigma_{YX} \Sigma_{XX}^{-1} (\mathbf{x} - \boldsymbol{\mu}_X) \right)^2$$

Step 3: Read off the result. This is the kernel of a univariate normal with mean $\mu_Y + \Sigma_{YX} \Sigma_{XX}^{-1} (\mathbf{x} - \boldsymbol{\mu}_X)$ and variance ω .

8.5 The same three properties hold

1. **Linear CEF.** $\mathbb{E}[Y \mid \mathbf{X} = \mathbf{x}] = \mu_Y + \Sigma_{YX} \Sigma_{XX}^{-1} (\mathbf{x} - \boldsymbol{\mu}_X)$ is linear in $\mathbf{x}$.
2. **Slope = regression coefficient.** The slope vector is $\boldsymbol{\beta} = \Sigma_{XX}^{-1} \Sigma_{XY} = [\text{Var}(\mathbf{X})]^{-1} \text{Cov}(\mathbf{X}, Y)$, which is the population multiple regression coefficient.
3. **Homoskedasticity.** The conditional variance $\sigma_Y^2 - \Sigma_{YX} \Sigma_{XX}^{-1} \Sigma_{XY}$ is a constant—it does not depend on $\mathbf{x}$.

The bottom line: nothing breaks in higher dimensions. Joint normality of $(Y, \mathbf{X})$ implies a linear CEF, normal errors, and homoskedasticity regardless of the number of regressors. The scalar fractions $\rho\sigma_Y/\sigma_X$ and $\sigma_Y^2(1-\rho^2)$ are simply replaced by their matrix counterparts $\Sigma_{YX}\Sigma_{XX}^{-1}$ and $\sigma_Y^2 - \Sigma_{YX}\Sigma_{XX}^{-1}\Sigma_{XY}$.

9 But doesn't the normal concentrate on a shell?

You may have encountered the following fact: if $\mathbf{Z} = (Z_1, \dots, Z_k)$ with $Z_j \stackrel{iid}{\sim} N(0, 1)$, then in high dimensions nearly all the probability mass lies near the *surface* of a sphere of radius $\sqrt{k}$, not near the origin. Specifically:

$$\|\mathbf{Z}\|^2 = \sum_{j=1}^k Z_j^2 \sim \chi_k^2, \quad \mathbb{E}[\|\mathbf{Z}\|^2] = k, \quad \text{Var}(\|\mathbf{Z}\|^2) = 2k$$

As k grows, the standard deviation $\sqrt{2k}$ becomes negligible relative to the mean k , so $\|\mathbf{Z}\|^2/k \rightarrow 1$ in probability. The mass concentrates in a thin shell at radius $\approx \sqrt{k}$.

This is real and important in high-dimensional statistics. But it does **not** undermine the regression result above, for two reasons.

9.1 The conditioning derivation is purely algebraic

The conditional distribution $Y \mid \mathbf{X} = \mathbf{x}$ was derived by dividing the joint density by the marginal density. This is a manipulation of density *functions*—it holds for every value of $\mathbf{x}$ in $\mathbb{R}^k$, regardless of how probable that particular $\mathbf{x}$ is. The formula

$$\mathbb{E}[Y \mid \mathbf{X} = \mathbf{x}] = \mu_Y + \Sigma_{YX}\Sigma_{XX}^{-1}(\mathbf{x} - \mu_X)$$

is an identity that holds at every $\mathbf{x}$, including points near the origin where mass is thin. Shell concentration tells us *where the data will be*, not that the formula changes at different locations.

9.2 What concentration *does* affect: finite-sample estimation

The population result is secure, but **estimating the regression in finite samples becomes genuinely harder** as k grows, and shell concentration is part of the reason. This deserves careful treatment, not a hand-wave.

The core problem: estimating Σ_{XX}^{-1} . The population regression slope is $\beta = \Sigma_{XX}^{-1}\Sigma_{XY}$. OLS estimates this with $\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$. Everything depends on how well the sample covariance matrix $\hat{\Sigma}_{XX} = \mathbf{X}'\mathbf{X}/n$ approximates Σ_{XX} . In low dimensions ($k \ll n$) this is reliable. In higher dimensions, several things go wrong:

1. **Parameter count outpaces data.** Σ_{XX} has $k(k+1)/2$ free parameters. With $k = 50$ regressors, that is 1,275 quantities to estimate. Even with $n = 200$ observations, each parameter is pinned down by very little data.
2. **Eigenvalue spreading (Marchenko–Pastur).** When k/n is not negligible, the eigenvalues of $\hat{\Sigma}_{XX}$ spread out relative to those of Σ_{XX} . The smallest sample eigenvalues are biased toward zero, the largest biased upward. Since $\hat{\Sigma}_{XX}^{-1}$ inverts these eigenvalues, low-variance directions get amplified, injecting noise into $\hat{\beta}$.
3. **Singularity when $k \geq n$.** If $k \geq n$, the sample covariance matrix is literally singular— $\mathbf{X}'\mathbf{X}$ has rank at most n , so $(\mathbf{X}'\mathbf{X})^{-1}$ does not exist and OLS cannot be computed at all.

Shell concentration is one geometric manifestation of these high-dimensional difficulties: when k is large, the norms $\|\mathbf{x}_i - \boldsymbol{\mu}_X\|$ concentrate near $\sqrt{k}$, which is a symptom of the same regime where $\hat{\Sigma}_{XX}$ becomes unreliable. But the central econometric issue is the instability of $\hat{\Sigma}_{XX}^{-1}$ when k is large relative to n , not shell concentration per se.

Consequences for OLS. Even though the population β is well-defined and the CEF is still linear, the OLS estimator in high dimensions can have:

- Enormous variance—individual $\hat{\beta}_j$'s bounce wildly across samples.
- Poor predictive performance— $\hat{Y} = \mathbf{x}'\hat{\beta}$ overfits the training data.
- Confidence intervals so wide as to be uninformative, or (worse) nominal coverage that is badly wrong because the t - and F -distributions assume k is fixed as $n \rightarrow \infty$.

What can be done. Regularization methods accept some bias in exchange for dramatically lower variance:

- **Ridge regression** shrinks $\hat{\beta}$ by replacing $(\mathbf{X}'\mathbf{X})^{-1}$ with $(\mathbf{X}'\mathbf{X} + \lambda\mathbf{I})^{-1}$, damping the noisy small-eigenvalue directions.
- **LASSO** adds an ℓ_1 penalty that drives some $\hat{\beta}_j$ to exactly zero, performing variable selection.
- **Principal component regression** projects $\mathbf{X}$ onto its leading eigenvectors before regressing, discarding the noisy directions entirely.

These methods implicitly acknowledge that the data do not contain enough information to estimate all k slopes reliably.

9.3 Other effects of high dimensions

Beyond the estimation of β , shell concentration creates additional difficulties:

- **Distance-based methods.** In high dimensions, pairwise distances become less informative, so methods that rely on finding “close” neighbors (e.g., kernel regression, k -NN) often deteriorate.
- **Density evaluation.** The joint density $f(\mathbf{x})$ evaluated at any particular point goes to zero as $k \rightarrow \infty$, even on the shell. This makes likelihood-based model comparison in very high dimensions difficult. The *conditional* density for the scalar Y given $\mathbf{X} = \mathbf{x}$ remains a well-behaved univariate normal, but evaluating the marginal likelihood requires integrating over the high-dimensional shell.

9.4 Bottom line

The population-level result is airtight: under joint normality, the CEF is linear with normal homoskedastic errors for any k . But the *estimation* problem degrades seriously as k grows relative to n , because the sample covariance matrix becomes a noisy estimate of Σ_{XX} and inverting it amplifies that noise. Shell concentration is one geometric symptom of this high-dimensional regime, not the root cause. The distinction matters—the *model* is fine, but our ability to *learn the model from data* is not, and recognizing this gap is the starting point for regularization and high-dimensional inference.

10 Summary

Starting from the bivariate normal density, we showed:

1. The conditional distribution $Y \mid X = x$ is normal (Section 4).
2. The conditional mean is linear in x , with slope $\rho\sigma_Y/\sigma_X = \text{Cov}(Y, X)/\text{Var}(X)$ (Section 5.1–5.2).
3. The conditional variance $\sigma_Y^2(1 - \rho^2)$ is constant in x (Section 5.3).

Under joint normality, the CEF *is* the population linear regression, and the classical normal regression assumptions hold by construction. Section 6 shows that the same result extends to k regressors via the partitioned multivariate normal, with scalar operations replaced by their matrix analogues. Section 7 explains why the concentration-on-a-shell phenomenon in high dimensions does not affect these results: the conditional distribution is derived pointwise for any $\mathbf{x}$, and the practical difficulties of high dimensions are estimation problems, not failures of the underlying population relationship.